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Acoustic Features of Music Differentially Modulate Anxiety through EEG Gamma Oscillations and Prefrontal Connectivity

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Abstract

University students may experience high levels of anxiety and depression, which can adversely affect their daily lives and academic performance and represent significant mental health conditions. Music therapy has been proposed as a promising noninvasive approach, but the role of specific acoustic features and their neural underpinnings remain unclear. In this study, we investigated the effects of tonality (major vs. minor) and timbre (piano vs. violin) on state anxiety. Seventy-four students were assigned to one of four intervention groups and passively listened to excerpts from Bach's works. Anxiety was assessed with the State-Trait Anxiety Inventory (STAI-S), and neural responses were recorded with EEG. Analyses included power spectral density of beta and gamma bands, EEG-based brain source localization analyses with standardized low resolution brain electromagnetic tomography (sLORETA), and Granger causality of directional connectivity. Results showed that major-key violin music elicited the

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strongest anxiety reduction among anxious participants. This effect was accompanied by gamma-band alterations, with Granger causality indicating that enhanced central-to-DLPFC connectivity predicted reductions in STAI-S scores, while dACC activation appeared as a concomitant effect. Beta and gamma oscillations further exhibited complementary properties, with gamma being more sensitive to music intervention and beta reflecting inter-individual stability. These findings will provide preliminary mechanistic insights into how elemental musical features can modulate anxiety.

Keywords: Anxiety, Music therapy, Violin, Major key, EEG, Gamma oscillations, Dorsolateral prefrontal cortex (DLPFC), sLORETA,

1. Background

State anxiety is a prevalent emotional state that, when left unmanaged, can develop into anxiety disorders. Such disorders not only impair cognitive functions such as attention and memory but also increase vulnerability to somatic conditions, including cardiovascular and immune dysfunction. (Reiche et al., 2004; Szuhany & Simon, 2022). The economic impact of anxiety is substantial, with annual healthcare costs and productivity losses estimated to surpass $4 billion (Harder et al., 2016). Therefore, early intervention for individuals experiencing anxiety is imperative to prevent its progression into a disorder. University students, with their heightened awareness of personal, interpersonal, and sociocultural differences, coupled with their openness to change, are particularly susceptible to psychological issues that can significantly impair cognitive functions such as thinking, perception, and learning (Endler & Kocovski,

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2001; Kocsis, 2013). Addressing anxiety among university students is thus crucial for their academic success and overall well-being.

Pharmacotherapy (PT) and cognitive–behavioral therapy (CBT) remain the most widely used and evidence-based approaches for the treatment of anxiety disorders (Agarwal & Landon, 2019; Chessick et al., 1996; Hans & Hiller, 2013; Watts et al., 2015). Both modalities have demonstrated substantial clinical benefits across different populations. Nonetheless, certain challenges persist: pharmacological treatments may be accompanied by side effects, particularly with long-term use, while CBT can show limited response in some individuals and relapse rates after discontinuation are not uncommon (Ginsburg et al., 2014; Swain et al., 2013; Weich et al., 2014). These limitations do not diminish the importance of PT and CBT, but they highlight the value of exploring complementary strategies that might provide additional options for specific groups of patients.

Music therapy (MT) is typically delivered by credentialed professionals within a therapeutic relationship in clinical practice, either in one-to-one or group settings over multiple sessions (Bradt et al., 2016). Such interventions often employ familiar or specially designed music and have shown efficacy in reducing anxiety, depression, and pain (Tang et al., 2021; Zhu et al., 2023). However, few studies have focused on younger populations, and the specific acoustic features and neural mechanisms underlying these effects remain insufficiently explored (Lu et al., 2021; Nilsson, 2008). These gaps highlight the need for experimental studies that systematically examine how specific musical features influence anxiety, particularly in younger populations.

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A wide range of musical features have been studied in various research contexts. Bach's classical music is often used because of its consistent style and therapeutic effectiveness (Sharda et al., 2019). Tonality plays a pivotal role in conveying emotions, with major and minor modes typically associated with feelings of happiness and sadness, respectively, due to their characteristic intervals (Balkwill & Thompson, 1999; Nieminen et al., 2012). The distinct timbres of the piano and violin also contribute to their differential effects on anxiety. The piano offers a balanced and wide harmonic spectrum, whereas the violin produces rich higher harmonics through bow friction (Sethares, 2005). On the basis of these findings, we have hypothesized that a specific musical element may be more effective in influencing anxiety states.

Emotion regulation of anxiety is supported by a distributed cortical network, including the prefrontal cortex (PFC), anterior cingulate cortex (ACC), insula, and orbitofrontal cortex (OFC). Within this network, the dorsolateral prefrontal cortex (DLPFC) is critically involved in cognitive control of emotion (Buhle et al., 2014; Ochsner et al., 2012), the ventrolateral and dorsomedial PFC (VLPFC, dmPFC) contribute to emotion inhibition and conflict processing(Kohn et al., 2014; Kuzmanovic et al., 2018), the dorsal ACC (dACC) monitors affective conflict and anxiety(A. Etkin et al., 2011; Fox et al., 2012), while the insula and OFC are central to interoception and affective evaluation(Craig, 2009; Rudebeck & Rich, 2018; Uddin et al., 2017) . Although these regions collectively shape emotional responses, the specific cortical substrates through which music exerts anxiolytic effects remain poorly understood.

The present study investigated how different musical instruments (piano vs. violin) and

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tonalities (major vs. minor) influence anxiety relief in university students. Participants listened passively to pre-selected excerpts from Bach Werke Verzeichnis (BWV), while subjective anxiety was assessed using the STAI-S and neural responses were recorded with EEG. Analyses included spectral power of beta and gamma bands, brain source localization using standardized low resolution brain electromagnetic tomography (sLORETA), and acoustic characterization of the music. To further probe mechanisms, we applied Granger causality and ANCOVA, providing an integrated evaluation of how specific musical elements modulate anxiety at both behavioral and neural levels. Taken together, these approaches offer an initial, controlled characterization of associations between specific musical features and anxiety-related measures in young adults and delineate hypotheses for future confirmatory studies.

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2. Methods and materials

2.1 Participants

Table 1. Demographic data of participants recruited for the anxiety group and normal control group.

	Anxiety group (n = 39)				Normal control (n = 34)			
	PA (n=9)	PI (n=11)	VA (n=11)	VI (n=8)	PA (n=8)	PI (n=8)	VA (n=9)	VI (n=9)
Age	24.47 (3.86)				23.36 (1.79)			
Female gender	26				18			
Male gender	12				12			
SAS scores	57.10 (8.54)				36.4 (5.40)			

All participants provided written informed consent, which was approved by the Ethics

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Committee of Biology and Medicine at Northwestern Polytechnical University in China. All methods were performed in accordance with the relevant guidelines and regulations. A total of 74 undergraduate and graduate students from Northwestern Polytechnical University in China were recruited through advertisements that invited students who perceived themselves as recently experiencing anxiety symptoms. All volunteers completed the Self-Rating Anxiety Scale (Zung, 1971) to provide a standardized assessment of anxiety severity, with scores ≥50 were classified into the “anxiety group,” while those scoring <50 were assigned to the “control group. Additional inclusion criteria were the absence of any other psychiatric disorders or familial psychiatric history, and no prior professional musical training. All the participants were subsequently randomly assigned to one of four groups (PA, PI, VA, VI), where "P" represents the piano group, "V" signifies the violin group, "A" denotes the major key, and "I" indicates the minor key, to receive various musical interventions. In total, the anxiety group included 26 females and 12 males (with a sample size: PA: 9, PI: 11, VA: 11 VI: 8), and the control group included 18 females and 12 males (PA: 8, PI: 8, VA: 9, VI: 9) (Table 1).

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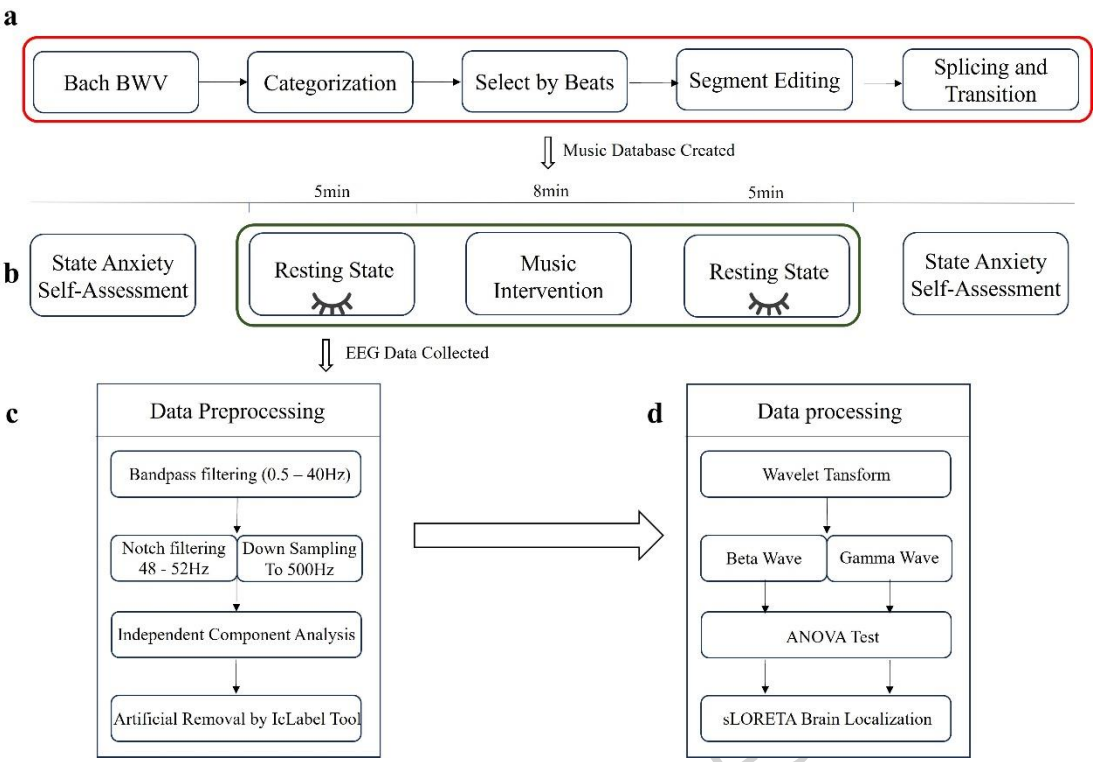

Fig 1. Experimental Design and Procedure for the EEG Experiment. **a**. Steps for processing music materials; **b**. Overall experiment flowchart; **c**. EEG data preprocessing process. **d**. EEG data analysis process. Abbreviations: EEG, electroencephalogram. BWV, Bach Werke Verzeichnis

Table 2. Beat-based Selected Music Library List and Random Splicing Results

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No.	Title	Performer	Category	Tonality	BMV	Start min	Duration (min)	Tempo(BPM/Hz)	Connected (Major)	Connected (Minor)
1	Sonata No.1 in G minor BWV 1001# I- Adagio	Itzhak Perlman	violin family	g minor	1001#1	0:00	1:39	62		
2	Partita No.1 in B minor BWV 1002# I- Allemanda	Itzhak Perlman	violin family	b minor	1002#1	2:35	1:31	66		
3	Partita No.2 in D minor BWV 1004#III- Sarabanda	Itzhak Perlman	violin family	d minor	1004#3	1:32	1:07	63		
4	Partita No.3 in E BWV 1006#II- Loure	Itzhak Perlman	violin family	E major	1006#2	0:00	1:47	78		
5	Sonata No.3 in C BWV 1005#II- Fuga	Itzhak Perlman	violin family	C major	1005#2	0:00	1:57	72		
6	Sonata No.3 in C BWV 1005#III- Largo	Itzhak Perlman	violin family	C major	1005#3	0:00	1:41	66		
7	Unaccompanied Cello Suite No. 2 in D minor, BWV 1008#2 - Prélude	MA Youyou	violin family	d minor	1008#2	0:00	1:53	65	4+5+6+8+9	1+2+3+7+10
8	Unaccompanied Cello Suite No. 1 in G Major, BWV 1007 - Sarabande	MA Youyou	violin family	G major	1007#1	0:00	1:33	63		
9	Unaccompanied Cello Suite No. 6 in D Major, BWV 1012 - Allemande	MA Youyou	violin family	D major	1012#1	0:00	1:44	62		
10	Unaccompanied Cello Suite No. 5 in C minor, BWV 1011	MA Youyou	violin family	c minor	1011#1	0:00	1:28	67		
1	Le Clavier bien tempéré - Livre 2: Prélude No.5 en Ré Majeur, BWV 874#5	ZHU Xiaomei	piano	D major	874#5	0:00	1:27	69		
2	Fugue No.8 in E-flat minor, BWV 853	ZHU Xiaomei	piano	e flat minor	853#8	0:00	1:25	90		
3	Goldberg Variations, BWV 988# Aria	ZHU Xiaomei	piano	G major	988	1:03	2:05	57		
4	Prélude N° 8 en Mi bémol mineur BWV 853	ZHU Xiaomei	piano	e flat minor	853#8	0:00	1:31	55		
5	Fugue N° 23 en Si majeur BWV 868	ZHU Xiaomei	piano	B major	868#23	0:00	1:59	73	1+3+5+6+9	2+4+7+8+10
6	Fugue N° 1 en Do majeur Bwv 846	ZHU Xiaomei	piano	D major	846#1	0:00	1:50	69		
7	Fugue N° 4 en Do dièse mineur Bwv 849	ZHU Xiaomei	piano	c flat minor	849#4	0:00	1:39	67		
8	Fugue N° 6 en Ré mineur Bwv 851	ZHU Xiaomei	piano	d minor	851#6	0:00	1:39	63		
9	Prélude N° 7 en Mi bémol majeur Bwv 852	ZHU Xiaomei	piano	E flat major	852#7	0:33	1:31	80		
10	Johann Sebastian Bach - Prélude N° 8 en Mi bémol mineur Bwv 853	ZHU Xiaomei	piano	e flat minor	853#8	0:35	1:39	60		

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2.2 Stimuli

The musical intervention materials utilized in the experiment included meticulously selected and edited instrumental music excerpts sourced from Bach's Works (BWV). As illustrated in Figure 1**a**, the entire selection and processing procedure encompassed six distinct stages: the establishment of a comprehensive music database, the categorization of musical materials, selection on the basis of tempo, the editing of segments, splicing and transitions, and final completion.

To minimize subjective emotional fluctuations and isolate the effects of specific musical elements (timbre: piano vs. violin; tonality: major vs. minor), we constructed a music database from Bach's works (BWV 772–994, 1001–1040), which are characterized by relatively uniform structures and limited dramatic progressions. The pieces were first categorized into four groups (PA, PI, VA, VI) and filtered to tempos of 60–120 beats per minute (BPM). From each group, five excerpts of approximately 1–2 minutes were chosen according to three criteria: (i) comparable phrase length, (ii) smooth stylistic continuity when concatenated, and (iii) avoidance of highly familiar pieces. Instead of using complete compositions, short phrases encompassing cadences were extracted and spliced in random order, producing final tracks that preserved tonal and timbral properties while reducing the influence of large-scale emotional trajectories (Table 2). The volume of all experimental tracks will be fixed to the same level.

2.3 Experimental Design

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2.3.1 STAI-S

In this study, the State Anxiety Subscale of the State-Trait Anxiety Inventory (STAI-S) was employed to explore the participants' subjective feelings and anxious state. Specifically, the STAI-S was utilized exclusively to assess and compare short-term anxiety states, without addressing the more enduring concept of "trait anxiety."

The subscale uses a 4-point Likert scale, where participants are required to rate the intensity of their feelings on a scale ranging from 1 (not at all) to 4 (very much so). The total score on the subscale can range from 20--80, with higher scores indicating a greater level of state anxiety. A score of 40 or above is generally considered indicative of high anxiety levels.

2.3.2 Experimental process

During the state anxiety intervention phase, including PA, PI, VA, and VI, participants were instructed to keep their phones on silent mode throughout the experiment to ensure a comfortable and uninterrupted experience. Furthermore, the experiment was conducted in a quiet, well-lit, and temperature-controlled (20–22°C) music therapy room with the use of headphones.

The entire experiment comprised five phases: an initial 5-minute resting period with eyes closed, an 8-minute state anxiety intervention, and a subsequent 5-minute resting period with eyes closed. Participants completed the State Anxiety Subscale of the State-Trait Anxiety Inventory (STAI-S) before (BF) and after (AF) the experiment, as illustrated in Figure 1**b**. Resting-state EEG segments from the BF and AF periods were

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collected for further analysis. Thus, each participant served as his/her own baseline control, with comparisons made between pre- and post-intervention resting states.

2.4 EEG Data Acquisition

EEG data were recorded from the subjects during both the resting state and the intervention state, with instructions to remain still throughout the recording process. The experimental data were collected via a 32-channel EEG system, model 8102, manufactured by Delica Medical Equipment Co., Ltd. in Shenzhen, China (Shenzhen Delica). Wet electrodes were utilized for the recording, with FCz serving as the reference electrode for online recording. The electrode cap layout adhered strictly to the international 10–20 system standard, and the sampling rate was set at 1000 Hz. Throughout the experiment, the impedance of all electrodes was maintained below 5 kΩ to ensure the accuracy and reliability of the collected data.

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2.5 Data analysis

2.5.1 Experimental Tracks

To ascertain the distinctions in spectral distribution between the timbre of a piano and a violin, the experimental audio was analyzed spectrographically via the fast Fourier transform (FFT) method, which specifically targeted the harmonic waveforms (Equation *2.1*):

$$X[f] = \sum_{t=0}^{N-1} x[t] \cdot e^{-j\frac{2\pi}{N}ft} \qquad (2.1)$$

where $X[t]$ is the discrete audio signal at time point t, N is the total number of

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sampled points, f denotes the frequency index, and $X[f]$ is the complex spectral coefficient at frequency f. The fundamental frequency was identified as the f at which $|X[f]|$ reached its maximum, and the corresponding harmonic magnitudes were subsequently calculated. For clarity, spectral magnitudes were averaged in 10 Hz bins and expressed in decibels (dB) for comparison. These spectral representations allowed us to compare the distribution of energy across frequencies between piano and violin tracks. Ultimately, the spectral graphs were generated via Matplotlib.

The roughness of each experimental audio piece was determined via MATLAB 9.13.0, where the audio signal was segmented into frames. The formula for calculating roughness, which is based on a psychoacoustic model proposed by Daniel and Weber (1997), is provided below (Equation *2.2*):

$$R = cal\sum_{i=1}^{47} (g(z_i) \cdot m_i^* \cdot k_{i-2} \cdot k_i)^2 [asper] \qquad (2.2)$$

In this formula, R represents the perceived roughness of the entire audio signal (unit: asper); z_i denotes the Bark-scale position of the i-th frequency band; $g(z_i)$ is the corresponding weighting function related to the bark scale (Aures, 1985); m_i^* is the effective modulation depth of band I; k_i and k_{i-2} represent specific loudness values used to capture inter-band interactions. The product term captures how modulation depth and loudness interact across adjacent bands, while the square ensures that contributions grow with energy. By summing across all critical bands, the model approximates how beating and fast amplitude fluctuations in the spectrum are integrated into the global percept of roughness.

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2.5.2 Preprocessing EEG Data

EEG data were preprocessed using the EEGLAB toolbox (Delorme & Makeig, 2004) implemented in MATLAB 9.13.0. A common average reference (CAR) method was applied, and a finite impulse response (FIR) filter was used to perform bandpass filtering between 0.5-80 Hz and notch filtering between 48-52 Hz. Additionally, the data were down-sampled from 1000 Hz to 500 Hz to improve computational efficiency and reduce data size. Artifact rejection was performed with the ICLabel plugin (Pion-Tonachini et al., 2019), which employs a pre-trained neural network classifier to assign probability values across seven categories (Brain, Muscle, Eye, Heart, Line noise, Channel noise, and Other). Independent components (ICs) with ≥80% probability of being Eye or Muscle artifacts were removed from the dataset. On average, approximately 0-3 ICs were rejected per participant (≈8–12% of total components), while the majority of the data (>85%) was retained after preprocessing (Figure 1**c**).

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2.5.3 EEG Data – Spectral Density Value Analysis

Analyses focused on beta (14–30 Hz) and gamma (30–80 Hz) activity at prefrontal (F3/F4) and central (C3/C4) electrodes, as these sites are commonly employed in studies of emotion regulation and anxiety-related brain activity. The continuous wavelet transform (CWT) was employed to derive the absolute power spectral values across all channels for all the subjects (Figure 1**d**). The wavelet basis function can be mathematically expressed as follows (Equation *2.3*):

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\psi\left(\frac{t-b}{a}\right) \qquad (2.3)$$

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where a is the scaling factor controlling frequency resolution, b is the translation parameter determining time localization, and t is the time variable. However, defining the wavelet basis alone does not yet describe how the EEG signal is analyzed. Therefore, the actual CWT is obtained by projecting the signal $x(t)$ onto this family of basis functions (Equation *2.4*):

$$WT_x(a,b) = \langle x(t), \psi_{a,b}(t) \rangle = \int x(t)\psi_{ab}{}^*(t)dt = \frac{1}{\sqrt{a}}\int_{-\infty}^{+\infty} x(t)\psi^*(\frac{t-b}{a})dt \quad (2.4)$$

This step extends the simple definition in *(2.3)* into a full transformation, enabling us to track how spectral power evolves across time. In this study, we applied the CWT to extract absolute power during the S1, T, and S2 periods, focusing on beta (14–30 Hz), and gamma (30–80 Hz) bands (Figure 1**d**).

2.5.4 EEG Data – Brain Source Localization Analysis

sLORETA-KEY is a commonly used algorithm for estimating cortical sources from low-density EEG, providing standardized solutions that reduce localization bias though with limited spatial resolution (Pascual-Marqui et al., 2018). In this study, we applied sLORETA to resting-state EEG recordings before and after each intervention to examine beta- and gamma-band activity. Signals were truncated to a 1-s epoch (150–151 s) from the middle of each resting period to avoid transitional artifacts and reduce computational load. After down-sampling to 500 Hz, the central 250–350 sampling points were analyzed. Pre–post differences were tested using paired-sample t-tests with permutation-derived thresholds, where voxels exceeding the permutation-derived critical *t* threshold were considered significant. Significant voxels were mapped to MNI

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coordinates and reported by Brodmann areas (BA).

2.5.5 Granger Causality Analysis

To assess the directional interactions between cortical regions, we applied Granger Causality (GC) analysis (Granger, 1969) to pre- and post-intervention EEG data. Directed functional connectivity was estimated in both the beta (12–30 Hz) and gamma (30–80 Hz) frequency bands. Specifically, GC analysis was performed at the electrode level, focusing on prefrontal (F3/F4; approximating DLPFC) and central (C3/C4) sites. Due to the medial location of the dACC and the limited spatial resolution of the 32/64-channel EEG montage, dACC signals could not be directly isolated for GC. From these estimates, three composite metrics were derived at the subject level: (i) DLPFC-to-Central, reflecting top-down prefrontal influences; (ii) Central-to-DLPFC, reflecting bottom-up cortical inputs to the DLPFC; and (iii) a Direction Index (DI), defined as the normalized difference between the two directions.

To examine the relationship with behavioral outcomes, GC indices were correlated with post-intervention STAI scores using both Spearman and Pearson correlation tests. In addition, multiple regression models (Y ~ X + STAIpre + group + music) were fitted to predict post-intervention STAI from GC metrics while controlling baseline anxiety, experimental group, and music condition. False discovery rate (FDR) correction was applied across tests to control for multiple comparisons.

2.6 Statistical analysis

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Statistical analyses included both within-subject and between-group comparisons. Within-subject changes (baseline vs. post-intervention resting) were assessed using paired t-tests on STAI scores and EEG power spectral values. Between-group effects of Group (Anxiety vs. Control), Condition (PA, PI, VA, VI), and their interaction were tested using two-way ANOVA on percentage decreases (Δ%). When the interaction was significant, simple-effects analyses were conducted; when normality or homogeneity assumptions were violated, Welch's ANOVA or non-parametric tests (Kruskal–Wallis with Dunn's post hoc = 6) were used. Assumptions were checked using Shapiro–Wilk and Levene's tests, complemented by visual inspection of residual distributions.

For source localization, sLORETA results were analyzed with paired t-tests and 5,000-permutation nonparametric tests, providing family-wise error rate (FWER) correction across 6239 voxels. Outliers were defined as observations with absolute z-scores > 3. To account for gender imbalance, an ANCOVA was performed with post-intervention scores as the dependent variable, music condition as the independent variable, and gender together with pre-intervention scores as covariates. All analyses were performed in GraphPad Prism (version 10.1.2). Statistical significance was set at $p < 0.05$.

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3. Results

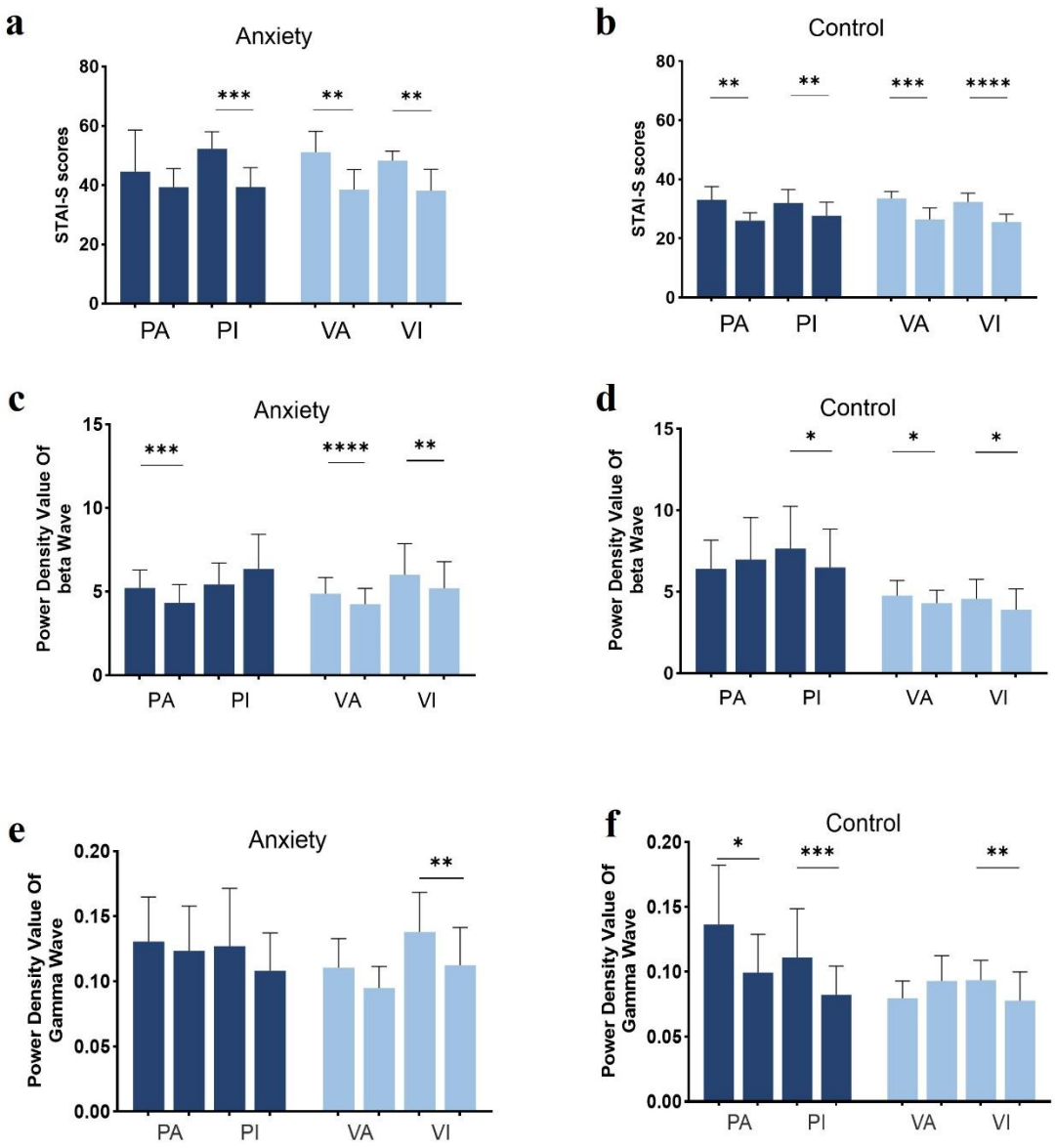

Fig 2. Effects of Music Interventions on STAI-S Scores and Brain Wave Power Density in Anxiety and Control Groups. (**a, b**) Changes in the STAI-S scores before and after the intervention in the anxiety group (**a**) and control group (**b**). (**c, d**) Power density of gamma waves in the anxiety group (**c**) and control group (**d**). (**e, f**) Power density of beta waves in the anxiety group (**e**) and control group (**f**). * P < 0.05, ** *P* < 0.01, *** *P* < 0.001, **** *P* < 0.0001. Abbreviations: PA, piano major; PI, piano minor; VA, violin major; VI, violin minor; P, piano; V, violin; A, major; I, minor; BF, before; AF,

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after. (n = 8-11)

3.1 Effects of Music Interventions on State Anxiety and EEG Power Spectra

To provide a preliminary reference for the subjective effectiveness of the interventions, STAI-S scores were collected before and after each session (Figure 2**a** and **b**). Across both anxiety (PI: $t(10) = 5.34$, $p = 0.0003$, Cohen's dz = 1.61; VA: t(10) = 3.91, p = 0.0029, Cohen's dz = 1.18; VI: t(6) = 5.00, p = 0.0025, Cohen's dz = 1.89) and nonanxiety participants (PA: t(6) = 4.68, p = 0.0034, Cohen's dz = 1.77; PI: t(6) = 4.12, p = 0.0062, Cohen's dz = 1.56; VA: t(6) = 6.25, p = 0.0008, Cohen's dz = 2.36; VI: t(8) = 7.21, p < 0.0001, Cohen's dz = 2.40), the scores showed a general downward trend . To complement the questionnaire data, we further examined EEG spectral power of beta and gamma waves (Figure 2**c** – 2**f**). In the anxiety group, one-way ANOVA revealed a marginal main effect of music condition on beta power (F(3,93) = 2.72, P = 0.049, R^2 = 0.081), with post hoc tests indicating decreases in the PA, VA, and VI conditions. Gamma power showed a more robust effect (F(3,140) = 2.59, P = 0.0003, R^2 = 0.12), with the VA condition producing the most pronounced reduction compared with the other groups. In the control group, significant main effects were also observed for both beta (F(3,93) = 2.72, P = 0.049, R^2 = 0.081) and gamma power (F(3,124) = 2.41, P < 0.02, R^2 = 0.076). Post hoc analyses indicated that PI and VI conditions were associated with significant gamma reductions relative to the other conditions (Figure 2C - 2F). Importantly, after adjusting for gender and pre-intervention scores in ANCOVA, the main effects of music condition remained unchanged, and no significant effect of gender was observed (F(1,35)=3.38, p=0.075, partial η^2=0.001).

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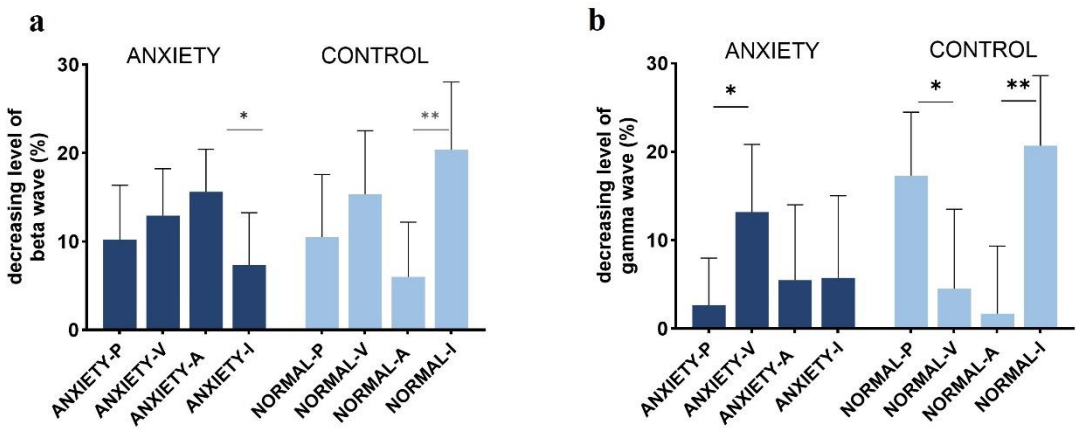

Fig 3. Changes in the spectral density of separated factors P, V, A, and I before and after music intervention in beta-brain waves (**a**) and gamma-brain waves (**b**). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. P = PA + PI; V = VA + VI; A = PA + VA; I = PI + VI. Abbreviations: PA, piano major; PI, piano minor; VA, violin major; VI, violin minor; P, piano; V, violin; A, major; I, minor.

Table 3. Changes in Power Spectral Density of Separated Factors P, V, A, and I before and after music interventions in Beta Brain Waves and Gamma Brain Wave

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	Group	Sample Size (N)	Effect Size (η^2)	Mean difference	Standard Deviation (SD)	t Value	Degrees of Freedom (df)	p Value
Beta Wave								
	ANXIETY-P	76	0.21	2.669	4.701	0.6628	149	0.5085
	ANXIETY-V	75						
	ANXIETY-A	79		-8.278	3.777	2.191	147	**0.03***
	ANXIETY-I	70						
	CONTROL-P	70	0.4	4.868	5.068	0.9605	130	0.3386
	CONTROL-V	62						
	CONTROL-A	75		14.38	4.918	2.923	170	**0.0041***
	CONTROL-I	67						
Gamma Wave								
	ANXIETY-P	72	0.44	10.53	4.684	2.248	141	**0.0261***
	ANXIETY-V	71						
	ANXIETY-A	73		0.0246	6.334	0.03887	148	0.969
	ANXIETY-I	77						
	CONTROL-P	69	0.23	-12.75	5.833	2.186	130	**0.0306***
	CONTROL-V	63						
	CONTROL-A	67		19	5.515	3.445	129	**0.0008***
	CONTROL-I	64						

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3.2 Differential Effects of Instrument and Tonality on Anxiety-Related Measures

To delve deeper into whether a single factor was responsible for the significant changes observed in the two-factor (timbre and tonality) groups, we isolated the four elements P, V, A, and I (where P = PA + PI, V = VA + VI, A = PA + VA, and I = PI + VI) and conducted a two-way ANOVA of EEG beta- and gamma-band percentage decreases (Δ%) (Figure 3**a** and **b**). A significant Group × Condition interaction was reported ($F = 4.343$ and $P = 0.00488$, $\eta^2 = 0.25$), with no significant main effects. Notably, in the Anxiety group, there was a significant anxiety-reduction effect caused by major music ($t(147) = 2.19$, $P = 0.030$, Cohen's $d = 0.36$) and violin music ($t(130) = 2.92$, $P = 0.0041$, Cohen's $d = 0.51$). Conversely, in the control group, minor music (A: $P = 0.0055$; B: $P = 0.0041$) and piano music ($t(141) = 2.25$, $P = 0.026$, Cohen's $d = 0.38$) led to a significant reduction in gamma waves (Table 3 & Figure 3). These findings suggest that major key violin music significantly reduces anxiety in individuals with high anxiety levels.

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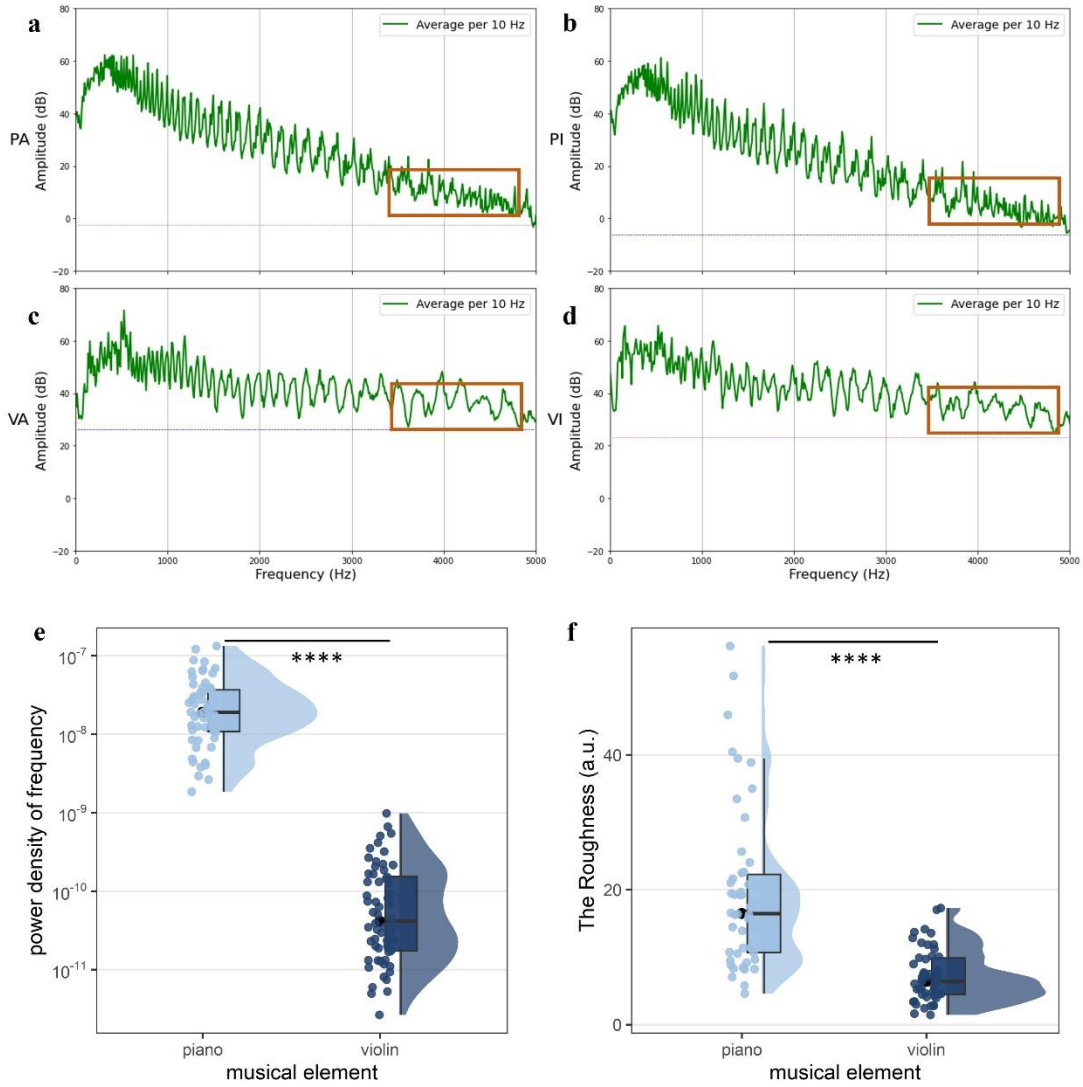

Fig 4. Spectral characteristics of experimental tracks.(**a-d**) Spectrograms of the PA (**a**), PI (**b**), VA (**c**), and VI (**d**) experimental tracks averaged every 10 Hz. (**e-f**) Raincloud plots depicting power density of frequency (**e**) and roughness (**f**) between piano and violin pieces in the 4000–5000 Hz range. Each plot shows the distribution, boxplot, and all individual data points. Violin music exhibited significantly greater power density (Hedges' g = 0.96) and roughness (Hedges' g = 1.17) than piano music. Negative values occasionally observed in kernel density estimation were artifacts of smoothing; raw data points confirmed all measurements were non-negative.

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3.3 Acoustic Characteristics of Violin and Piano Music: Frequency Density and Roughness

To gain a deeper understanding of how different musical elements influence anxiety levels, we employed the fast Fourier transform (FFT) to analyze the harmonic spectrum of each experimental track. Distinct waveform differences between the piano group (Figure 4**a** and **b**) and the violin group (Figure 4**c** and **d**) were evident in the highlighted squares based on the frequency density of high-frequency harmonics. To emphasize the significance of these differences, we conducted a statistical analysis of the roughness and power density of frequency on the experimental tracks. Notably, the violin group presented significantly greater roughness ($P < 0.0001$, Hedges' $g = 1.17$, 95% CI [0.68, 1.65]) and greater frequency power density ($P < 0.0001$, Hedges' $g = 0.96$, 95% CI [0.50, 1.41]) than did the piano group, suggesting that the acoustic characteristics of these two instruments in the high-frequency range are markedly distinct (Figure 4**e** and **f**).

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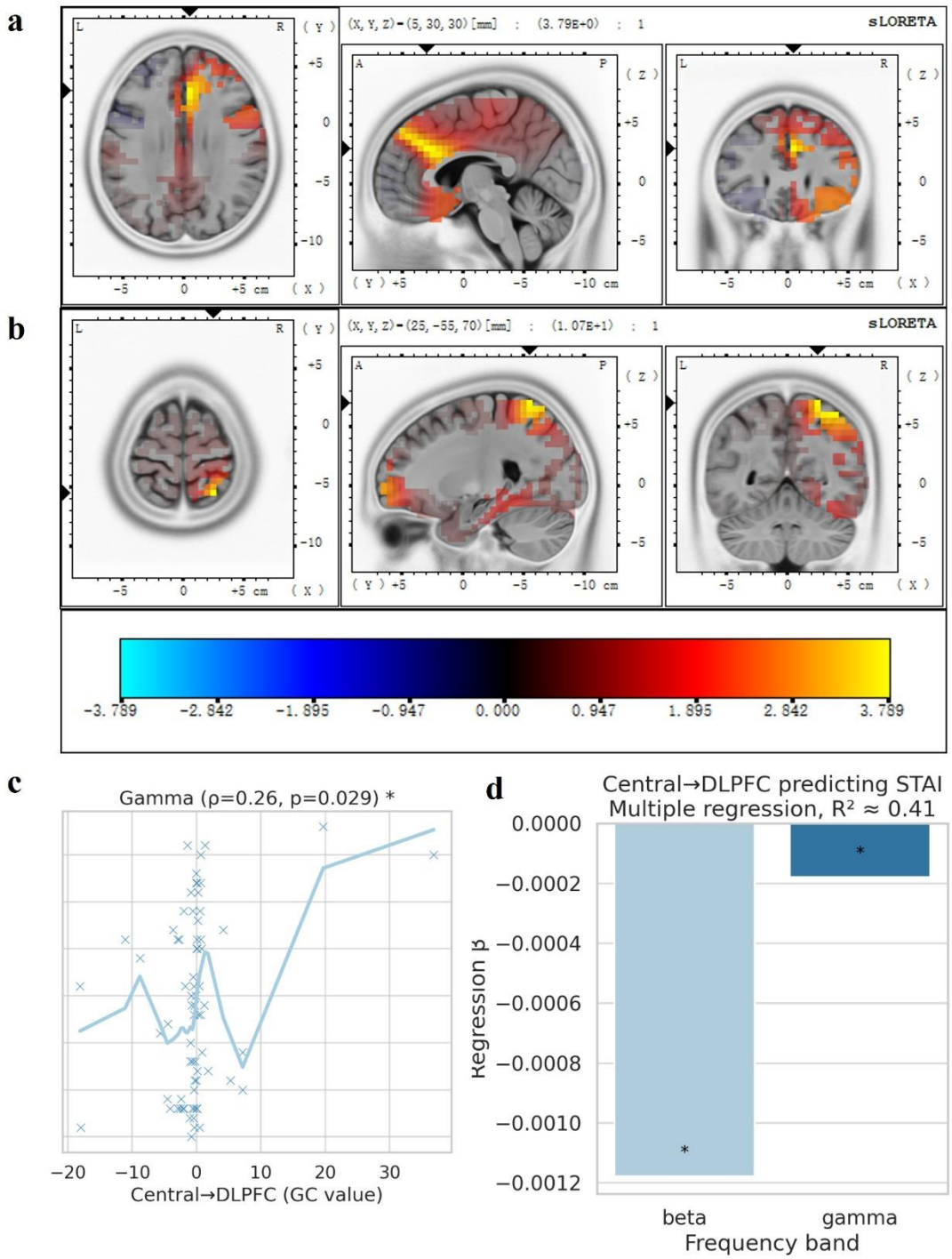

Fig 5. sLORETA and Granger causality results. (**a-b**) XYZ sLORETA slices of the average change in gamma waves in the anxiety group following intervention with VA music (**a**) and PI music (**b**); red voxels indicate regions with greater activation. Coordinates correspond to the most prominently activated locations. (**c**) Scatter plot showing the relationship between Central-to-DLPFC connectivity (GC values) and

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post-intervention STAI scores in the gamma band ($\rho = -0.26$, $p = 0.029$). (**d**) Bar plot illustrating multiple regression results, where Central-to-DLPFC connectivity in the gamma band significantly predicted post-intervention STAI scores after controlling for baseline STAI, group, and music condition ($R^2 \approx 0.41$). * $p < 0.05$. Abbreviations: PI, piano minor; VA, violin major; DLPFC, dorsolateral prefrontal cortex; STAI, State-Trait Anxiety Inventory.

Table 4. Brain regions showing maximal changes in beta- and gamma-band activity across musical interventions. Only the VA–gamma condition exhibit significant SnPM correction (P = 0.0376, coloured in red); other regions are reported descriptively.

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	Group		Broadman Area number	Specific Location
Beta Wave				
	Violin	Anxiety	BA 9, 32	Dorsolateral Prefrontal Cortex, Dorsal Anterior Cingulate Cortex
		Control	BA 40, 2	Supramarginal Gyrus, Primary Somatosensory Cortex
	Piano	Anxiety	BA 9, 10	Dorsolateral Prefrontal Cortex, Anterior Prefrontal Cortex
		Control	BA 9, 10	Dorsolateral Prefrontal Cortex, Anterior Prefrontal Cortex
	Major	Anxiety	BA 7, 40	Superior Parietal Lobule, Supramarginal Gyrus
		Control	BA 7, 31	Superior Parietal Lobule, Dorsal Posterior Cingulate Cortex
	Minor	Anxiety	BA 47#, 11#	Inferior Frontal Gyrus, Orbitofrontal Cortex
		Control	BA 40, 2	Supramarginal Gyrus, Primary Somatosensory Cortex
Gamma Wave				
	Violin	Anxiety	BA 9#, 32#	Premotor Cortex, Dorsolateral Prefrontal Cortex, Dorsal Anterior Cingulate Cortex
		Control	BA 23, 31, 10	Ventral Posterior Cingulate Cortex, Dorsal Posterior Cingulate Cortex, Anterior Prefrontal Cortex
	Piano	Anxiety	BA 13#, 47#, 38#	Insular Cortex, Inferior Frontal Gyrus, Temporal Pole
		Control	BA 32#, 24	Dorsal Anterior Cingulate Cortex, Ventral Anterior Cingulate Cortex
	Major	Anxiety	BA 32#	Dorsal Anterior Cingulate Cortex, Ventral Anterior Cingulate Cortex
		Control	BA 5, 7, 13#	Somatosensory Association Cortex, Superior Parietal Lobule, Insular Cortex
	Minor	Anxiety	BA 40, 2	Supramarginal Gyrus
		Control	BA 19#, 30*#	Visual Association Cortex, Retrosplenial Cortex

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3.4 Cortical Activity Patterns Associated With Major Violin Music

Given the anxiety-reducing effects of major violin music, we next examined pre–post resting-state changes in cortical current density using sLORETA. Permutation-corrected analyses (SnPM, 5,000 randomizations, family-wise error corrected) demonstrated that only the Major (A) and Violin (V) intervention in the gamma band in Anxiety group yielded significant pre–post changes (A: t = 3.40, P = 0.036; V: t = 2.92, P = 0.045). Significant clusters were localized to the dorsolateral prefrontal cortex, and dorsal anterior cingulate cortex (BA 9, 32) intervened by Violin factor, and similarly, to the anterior cingulate cortex (BA32) intervened by the Major factor (Table 4 and Figure 5**a** and **b**). Nevertheless, to provide a complete overview of the data, the regions showing the strongest uncorrected changes are reported descriptively in Table 4. These included, for example, the ventral and dorsal posterior cingulate cortices and anterior prefrontal cortex (BA 23, 31, 10) in the control group, the insular cortex, inferior frontal gyrus, and temporal pole (BA 13, 47, 38) following piano interventions in the anxiety group, and the supramarginal gyrus and primary somatosensory cortex (BA 40, 2) under minor-key interventions in the anxiety group.

3.5 Granger Causality between Central Cortex and DLPFC Predicts Anxiety Reduction

GC analysis was conducted to examine directional connectivity between central cortical sites and the DLPFC in the beta and gamma frequency bands. Central-to-DLPFC

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connectivity showed a significant negative correlation with post-intervention STAI scores in the gamma band ($\rho = -0.26$, $p = 0.029$; Fig. 5C). Multiple regression models further confirmed that Central-to-DLPFC connectivity was a significant predictor of post-intervention STAI scores after controlling for baseline STAI, group, and music condition ($\beta < 0$, $p < 0.05$, $R^2 \approx 0.41$; Fig. 5D).

4. Discussion

Music therapy has been increasingly recognized as a promising noninvasive approach for anxiety relief, yet the diversity of musical attributes and the complexity of neural mechanisms present challenges for systematic investigation. In this study, we investigated the independent effects of tonality (major *v.s*. minor) and timbre (piano *v.s* violin) on state anxiety, integrating behavioral outcomes with EEG spectral and source localization analyses. Among anxious individuals, major-key violin music showed the most pronounced anxiolytic effect, which was accompanied by gamma-band alterations and engagement of prefrontal circuits. While the observed dACC activation may represent a concomitant effect of anxiety reduction, Granger causality analysis provided stronger evidence that enhanced central-to-DLPFC connectivity in the gamma band was functionally linked to reductions in post-intervention STAI scores. Together, these results provide preliminary evidence that specific acoustic properties of music can engage prefrontal regulatory pathways, offering a cautious step toward clarifying how elemental musical features may influence anxiety.

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To ensure that the effects of specific musical elements could be examined under controlled conditions, we carefully standardized the music materials used in this study. We selected music by Bach, as Baroque compositions are known for their relatively uniform style and are less subject to the expressive fluctuations seen in other musical periods (Sharda et al., 2019). Additionally, we constructed the experimental track by splicing together several ordinary musical phrases rather than using complete pieces, which was done to avoid the emotional progression, climax, and structural development that a full composition typically contains (Fedorenko et al., 2012). Finally, the tempos to a specific BPM were constrained within the spontaneous tempo of human physiological rhythms (e.g., heart rate, locomotion) to minimize the physiological effects of rhythm on participants (MacDougall & Moore, 2005).

Music therapy has shown promise across various clinical settings (Tang et al., 2021). To validate its anxiety-reducing effects, this study recorded participants' STAI-S score data and EEG data before and after music interventions. On average, participants showed a significant reduction in STAI-S scores after the intervention. (Figure 2), indicating that the selected music database, which is characterized by its classical style and physiological-fitting rhythm, can generally possess anxiety-reducing properties, which aligns with prior research (Barlas et al., 2023; Weineck et al., 2022). However, since each condition (e.g., PA, VI) inherently combines two musical features, the current analysis cannot disentangle whether the observed changes are primarily attributable to timbre, tonality, or their interaction. To clarify the contribution of these individual elements, we therefore decomposed the conditions into their constitutive

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factors (P, V, A, I) and reanalyzed the data using percentage decreases as the outcome measure. This approach allows a more precise evaluation of how timbre and tonality independently and jointly shape neural responses associated with anxiety.

Tonality, a fundamental aspect of music, plays a pivotal role in emotional communication (Balkwill & Thompson, 1999), with major and minor keys often associated with happiness and sadness, respectively. Therefore, we compared Major (A) element and Minor (I) element to assess the impact of musical emotions on anxiety. Our analysis revealed that, in the control group, minor key music led to a greater reduction in anxiety than major key music. Conversely, anxious individuals showed greater anxiety reduction with major key music, experiencing a more pronounced anxiety-relieving effect (Figure 3**a**). This finding is consistent with the Mood Congruence Hypothesis (Millgram et al., 2015), which posits that individuals tend to preferentially select external stimuli congruent with their current emotional state. In terms of the psychological mechanism of this framework, Yoon et al (2020) further discussed the motivational basis of the Mood Congruence Hypothesis, suggesting that the essence of this phenomenon lies in emotion regulation needs guiding stimulus selection. For example, individuals experiencing sadness may choose low-energy music (i.e., sad music) to alleviate sadness and seek calmness, thereby achieving emotional soothing. This introduces the concept of the energy dimension, which refers to the arousal level of music. Anxiety is a high-energy emotion and major key music is considered high-energy because of its capacity to evoke heightened arousal (Juslin & Laukka, 2004; Punkanen et al., 2011; Ramirez & Vamvakousis, 2012). We speculate that the normal

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group showed greater anxiety reduction when listening to minor key music, possibly because its relatively low-energy characteristics induce calmness. In contrast, anxious individuals, with their high-energy emotional state, may have been more easily engaged or distracted by high-energy music such as major key pieces, which could partly explain their anxiety reduction. Besides distraction, another plausible mechanism is the induction of positive mood by music, which has been well documented in previous research (Juslin & Laukka, 2004; Koelsch, 2010; Salimpoor et al., 2011). Such mood induction effects could have contributed to the anxiety reduction observed in our study.

Timbre stands as a pivotal characteristic in the realm of music. To further examine how different timbres influence emotions, the impact of Piano (P) and Violin (V) music on anxiety relief was investigated. Intriguingly, the results revealed a discernible difference in the effects of instrument type between the anxious cohort and the normal control group. Specifically, violin music had a more potent anxiety-alleviating effect on the anxious group, whereas piano music was more efficacious for the control group, as illustrated in Figure 3**b**. The soothing effect of piano music on emotions and its stress-relieving capabilities have been widely reported (Moreno-Gutiérrez et al., 2023; Musa et al., 2022; Toyoshima et al., 2011; Yang, 2025; Yoshie et al., 2009). Conversely, the stronger anxiety-reducing effect of violin music among anxious individuals presents an unexpected twist.

To better understand the factors driving these divergent outcomes, we conducted a spectral analysis of the experimental tracks. The findings revealed that violin music exhibited higher frequency density and greater roughness as shown in Figure 4. Notably,

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higher spectral density increases the likelihood of closely spaced partials, which interact through beating and thereby give rise to auditory roughness (Vassilakis, 2001; Zwicker & Fastl, 2013). This provides a mechanistic link between the observed spectral properties of violin music and its elevated roughness levels. Interestingly, many studies have reported that auditory roughness is closely associated with aversive perception (Arnal et al., 2015; Arnal et al., 2019; Björk, 1985; Bodner et al., 2007; Koutseff et al., 2018; Štillová et al., 2021). However, it should be noted that, instruments have complex time-frequency structures, but their primary design is not intended to trigger threat-related responses. As a result, string instruments with higher levels of roughness do not elicit strong aversive reactions (Arnal et al., 2015). Therefore, the music used in this study can disregard the "unpleasantness" typically associated with roughness and instead focus on the direct impact of "roughness" itself. From a cognitive perspective, Taffou et al (2021) demonstrated that auditory roughness triggers defensive responses by enlarging the peripersonal space and priming the organism for threat-related behaviors. The so-called “defensive” responses to rough sounds can essentially be interpreted as heightened attentional allocation to these stimuli, which are biologically encoded as salient threat cues and exert strong attentional capture effects (Arnal et al., 2015). We therefore propose that auditory roughness may serve as a mechanism of distraction and attentional redirection, thereby contributing to the alleviation of anxiety symptoms.

To deeply explore the neurobehavioral relevance of rough music, we conducted source localization analyses on the EEG S1 and S2 data. The sLORETA results in gamma wave

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emphasized that interventions involving major keys and violin timbre produced significant cortical activation in the anxiety group, specifically engaging Brodmann areas 9 and 32 (Table 4 and Figure 5**a** and **b**). This finding extends the anxiolytic effect of major violin music to the level of neural responses. BA 9 and 32 are located in the frontal lobe, where the dorsal anterior cingulate cortex (dACC, BA 32) has been implicated in monitoring affective conflict and anxiety (Etkin et al., 2011), and the dorsolateral prefrontal cortex (DLPFC, BA 9) plays a central role in top-down emotion regulation by inhibiting limbic overactivation and reallocating attention toward external cues (Bishop, 2009; Ochsner & Gross, 2005). From a broader neural perspective on auditory roughness, Arnal et al. (2015) demonstrated that rough sounds strongly engage the amygdala, while subsequent work showed a cascade of responses - initial startle activity in the auditory cortex (high-gamma synchronization), followed by limbic activation (insula, amygdala), and finally recruitment of parietal and prefrontal cortices responsible for regulation and attentional control (Arnal et al., 2019). Taken together, the observed dACC and DLPFC activation in our sLORETA analyses aligns with established roughness-related cortical pathways, suggesting that the distinctive roughness of major violin music may recruit prefrontal circuits to support its anxiolytic effects.

To further verify whether the activation of the dACC and DLPFC was driven by music intervention rather than representing a concomitant phenomenon, we performed Granger causality (GC) analysis. Because the dACC is an intermediate structure that cannot be directly isolated with 32-channel EEG, its results were considered only as

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supportive evidence of prefrontal involvement based on source localization. Focusing on the central cortical sites and the DLPFC, the GC analysis revealed that in the gamma band, the Central-to-DLPFC directional connectivity was significantly negatively correlated with post-intervention STAI scores and significantly predicted anxiety reduction in regression models (Figure 5C). To further verify the robustness of this relationship, multiple regression analyses were conducted controlling for baseline STAI, group, and music condition, which confirmed that Central→DLPFC connectivity significantly predicted anxiety reduction (Figure 5D). These findings strengthen the interpretation that DLPFC engagement during VA intervention is not merely epiphenomenal but functionally linked to anxiety relief through enhanced bottom-up input. This aligns with classical models highlighting the role of the ACC–PFC circuit in emotion regulation (Etkin et al., 2015), and with more recent reviews emphasizing the critical involvement of DLPFC in controlling anxious states via its executive and attention-control functions (White et al., 2023).

Both beta and gamma bands obtained a comprehensive picture of oscillatory responses to music intervention (Figure 2 to Figure 5) but two rhythms exhibited distinct patterns: beta activity showed only marginal modulation by music condition, whereas gamma activity displayed more robust changes, particularly in the anxiety group. Granger causality analysis further suggested that gamma-band connectivity was more strongly associated with anxiety reduction, while beta effects were weaker and nonsignificant. This is consistent with previous evidence that gamma oscillations provide a sensitive index of emotion-cognition interactions (Yang et al., 2020) and may reflect rapid

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information transfer within the anterior cingulate-prefrontal circuit (Amit Etkin et al., 2011). By contrast, beta activity appeared to capture more general arousal- and attention-related processes. Interestingly, when baseline covariates were considered in ANCOVA, beta effects became more pronounced, suggesting that beta oscillations may provide a stable marker of inter-individual variability. Taken together, our findings indicate that gamma oscillations are more sensitive to music-induced modulation, whereas beta oscillations contribute complementary stability, highlighting the value of considering both bands for a more complete understanding of neural mechanisms underlying music-based anxiety relief.

Our paradigm differs from conventional clinical music therapy, where therapists typically use complete pieces with intact emotional trajectories to guide clients. In contrast, we intentionally minimized expressive elements to isolate the effects of specific acoustic features such as tonality and timbre under controlled conditions. Although this reductionist design cannot reproduce the ecological richness of therapist–client interactions, it provides useful preliminary insights. Understanding how isolated musical elements modulate neural activity may help refine therapeutic music selection and, in the future, contribute to AI-assisting tools that complement therapist-guided interventions with a deeper mechanistic basis.

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5. Limitations

To isolate the effects of specific acoustic features, we constrained tempo and selected short Baroque phrases for splicing, thereby minimizing large-scale emotional

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trajectories. While this design provided a controlled framework, it may also have weakened the ecological validity of the music, and the emotional valence of the final tracks was not independently validated. Moreover, subtle tempo differences and the lack of perfect key matching across groups could have introduced additional bias, and all excerpts were performed by only three musicians, leaving open the possibility of performer-specific effects. Future studies should therefore consider larger and standardized music libraries, as well as objective validation of musical features, to enhance reproducibility.

Another limitation concerns both the EEG data acquisition and sample size. Source localization was performed with a 32-channel EEG system, which inherently provides lower spatial resolution than high-density arrays; hence, the reported cortical activations should be interpreted with caution. Similarly, our Granger causality analyses were limited to electrode-level data and behavioral correlations, rather than allowing mechanistic inference about interregional connectivity. In addition, the relatively small sample size within each intervention subgroup reduced statistical power and may limit generalizability. Future work should aim to replicate these findings with larger samples, high-density or multimodal neuroimaging (e.g., fMRI–EEG), and more advanced causal modeling approaches such as dynamic causal modeling (DCM).

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6. Conclusion

In summary, this study systematically examined how tonality and timbre shape the anxiolytic effects of music in university students. Major-key violin music emerged as

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particularly effective in reducing state anxiety, accompanied by gamma-band modulation and engagement of prefrontal circuits. While dACC activation may represent a concomitant effect, Granger causality analyses indicated that enhanced central-to-DLPFC connectivity was functionally linked to anxiety relief. Moreover, gamma brain wave provided a more sensitive marker of music-induced modulation, whereas beta offered complementary stability when individual variability was accounted for. Though using decontextualized stimuli, this approach decodes hierarchical music-emotion circuitry interactions, advancing targeted acoustic therapy development.

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Declarations

Ethics approval statement

This study received approval from the Ethics Committee of Biology and Medicine at Northwestern Polytechnical University. (Approval No: 202302054)

Patient consent statement

All participants provided written informed consent and were given detailed information regarding the study. Appropriate compensation was provided, and the confidentiality of personal data was strictly maintained.

Data availability statement

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The datasets utilized and examined in this study can be obtained from the corresponding author upon reasonable request.

Conflict of interest disclosure

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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Authors' Contributions

Q. LUO was involved in conceptualization, methodology, software, formal analysis, investigation, writing, writing-review & editing, visualization, and funding acquisition. Y. WANG contributed to the experimental design, data processing, and manuscript preparation and revision. K. YANG contributed to data analysis, result visualization, and interpretation. W. ZHANG, M. XI, A. QIAN, and P. NIE provided resources. A. QIAN also participated in writing-review & editing and funding acquisition. G. CHEN was responsible for supervision. All authors read and approved the final manuscript.

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Figure Legends

Fig 1. Experimental Design and Procedure for the EEG Experiment. **a**. Steps for processing music materials; **b**. Overall experiment flowchart; **c**. EEG data preprocessing process. **d**. EEG data analysis process. Abbreviations: EEG, electroencephalogram. BWV, Bach Werke Verzeichnis

Fig 2. Effects of Music Interventions on STAI-S Scores and Brain Wave Power Density in Anxiety and Control Groups. (**a, b**) Changes in the STAI-S scores before and after the intervention in the anxiety group (**a**) and control group (**b**). (**c, d**) Power density of gamma waves in the anxiety group (**c**) and control group (**d**). (**e, f**) Power density of beta waves in the anxiety group (**e**) and control group (**f**). * P < 0.05, ** *P* < 0.01, *** *P* < 0.001, **** *P* < 0.0001. Abbreviations: PA, piano major; PI, piano minor; VA, violin major; VI, violin minor; P, piano; V, violin; A, major; I, minor; BF, before; AF, after. (n = 8-11)

Fig 3. Changes in the spectral density of separated factors P, V, A, and I before and after music intervention in beta-brain waves (**a**) and gamma-brain waves (**b**). * *P* < 0.05, ** *P* < 0.01, *** *P* < 0.001, **** *P*< 0.0001. P = PA + PI; V = VA + VI; A = PA + VA; I = PI + VI. Abbreviations: PA, piano major; PI, piano minor; VA, violin major; VI, violin minor; P, piano; V, violin; A, major; I, minor.

Fig 4. Spectral characteristics of experimental tracks.(**a-d**) Spectrograms of the PA (**a**),

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PI (**b**), VA (**c**), and VI (**d**) experimental tracks averaged every 10 Hz. (**e-f**) Raincloud plots depicting power density of frequency (**e**) and roughness (**f**) between piano and violin pieces in the 4000–5000 Hz range. Each plot shows the distribution, boxplot, and all individual data points. Violin music exhibited significantly greater power density (Hedges' g = 0.96) and roughness (Hedges' g = 1.17) than piano music. Negative values occasionally observed in kernel density estimation were artifacts of smoothing; raw data points confirmed all measurements were non-negative.

Fig 5. sLORETA and Granger causality results. (**a-b**) XYZ sLORETA slices of the average change in gamma waves in the anxiety group following intervention with VA music (**a**) and PI music (**b**); red voxels indicate regions with greater activation. Coordinates correspond to the most prominently activated locations. (**c**) Scatter plot showing the relationship between Central-to-DLPFC connectivity (GC values) and post-intervention STAI scores in the gamma band ($\rho = -0.26$, $p = 0.029$). (**d**) Bar plot illustrating multiple regression results, where Central-to-DLPFC connectivity in the gamma band significantly predicted post-intervention STAI scores after controlling for baseline STAI, group, and music condition ($R^2 \approx 0.41$). * $p < 0.05$. Abbreviations: PI, piano minor; VA, violin major; DLPFC, dorsolateral prefrontal cortex; STAI, State-Trait Anxiety Inventory.

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Figure 1

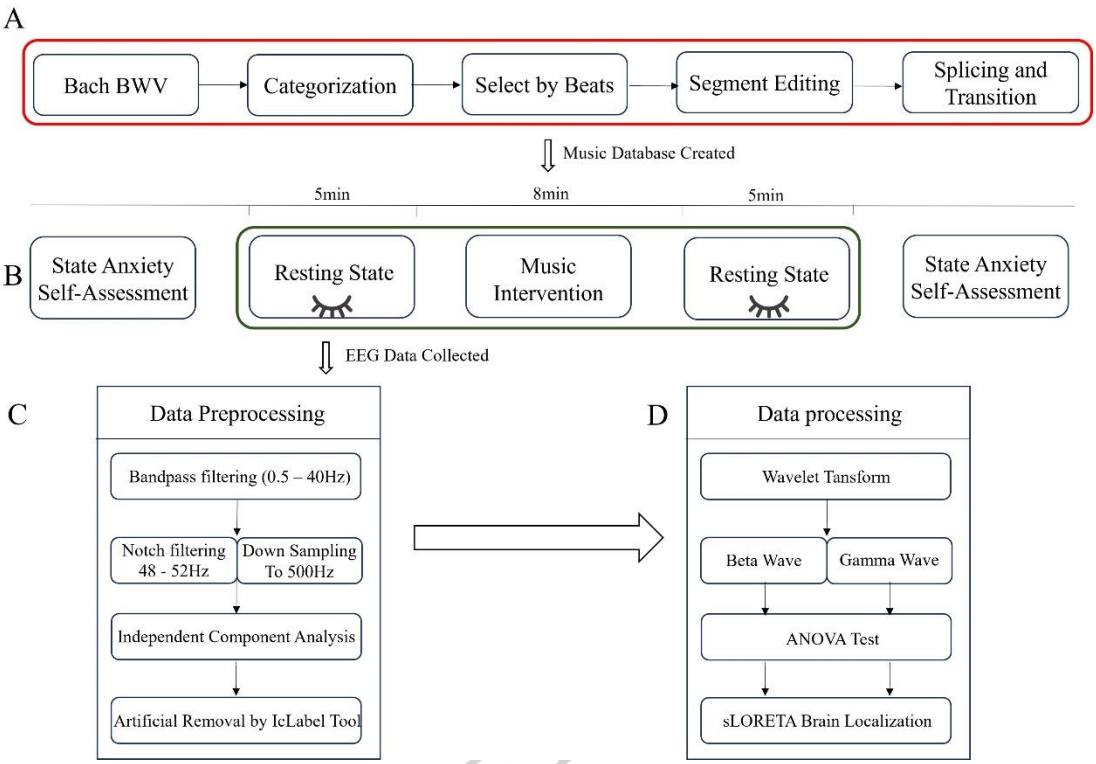

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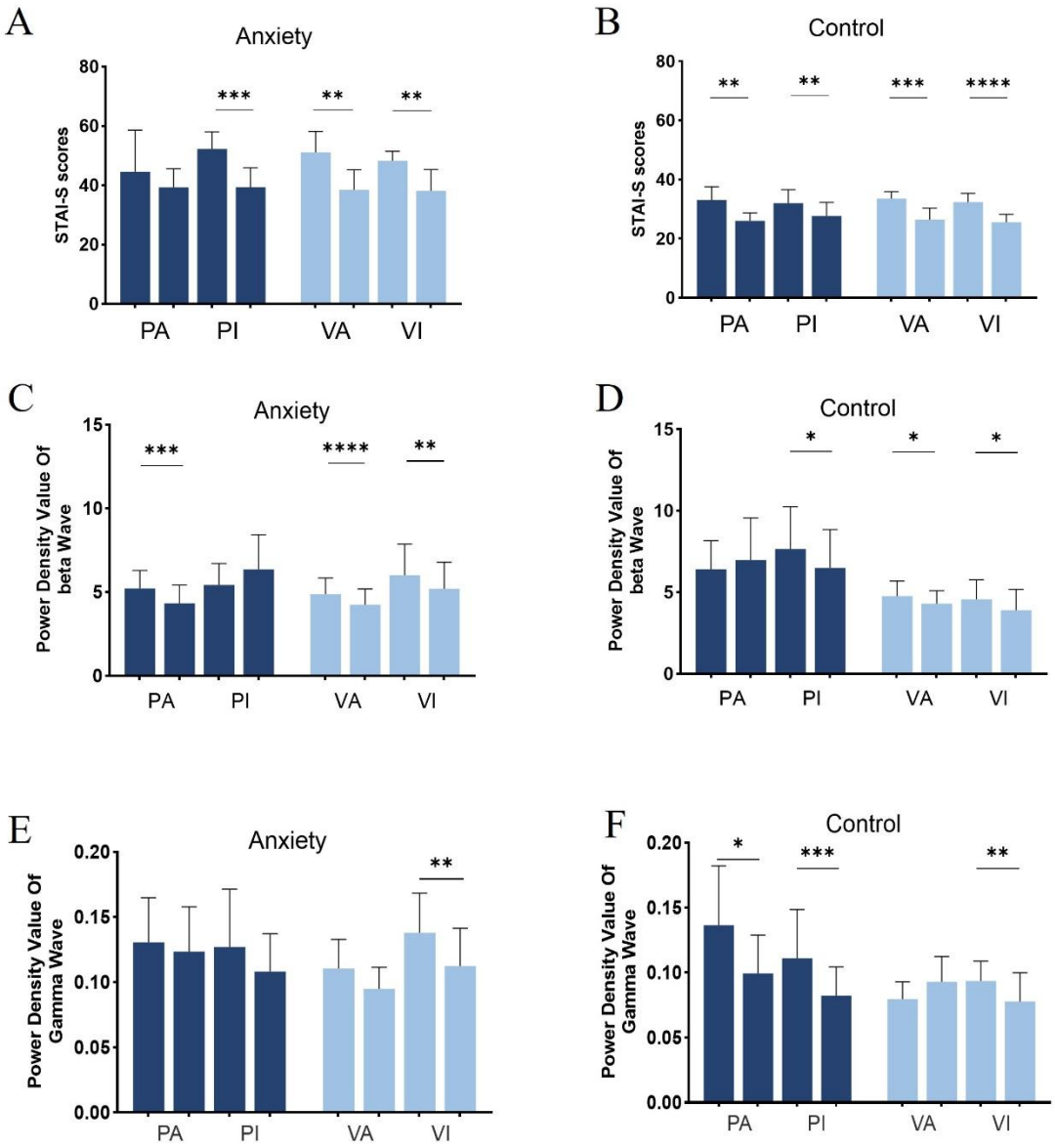
Figure 2


A
Anxiety
STAI-S scores
80
60
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**
**
PA
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VA
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B
Control
STAI-S scores
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40
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Anxiety
Power Density Value Of beta Wave
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Anxiety
Power Density Value Of Gamma Wave
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Figure 3

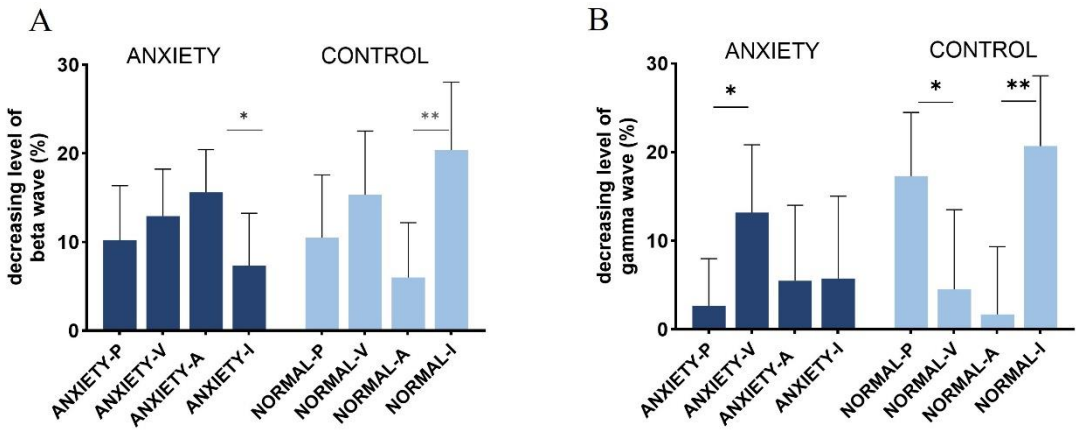

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Figure 4

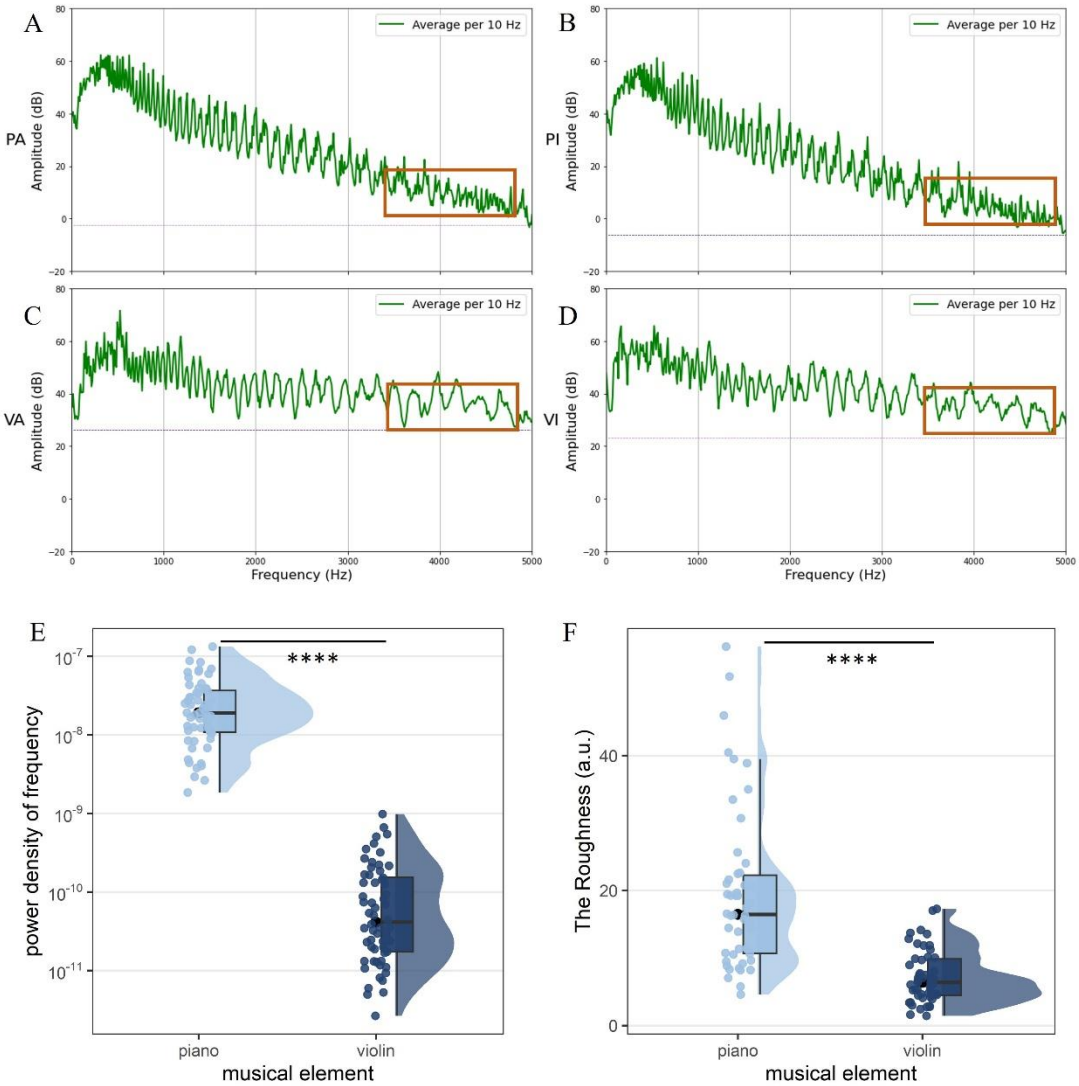
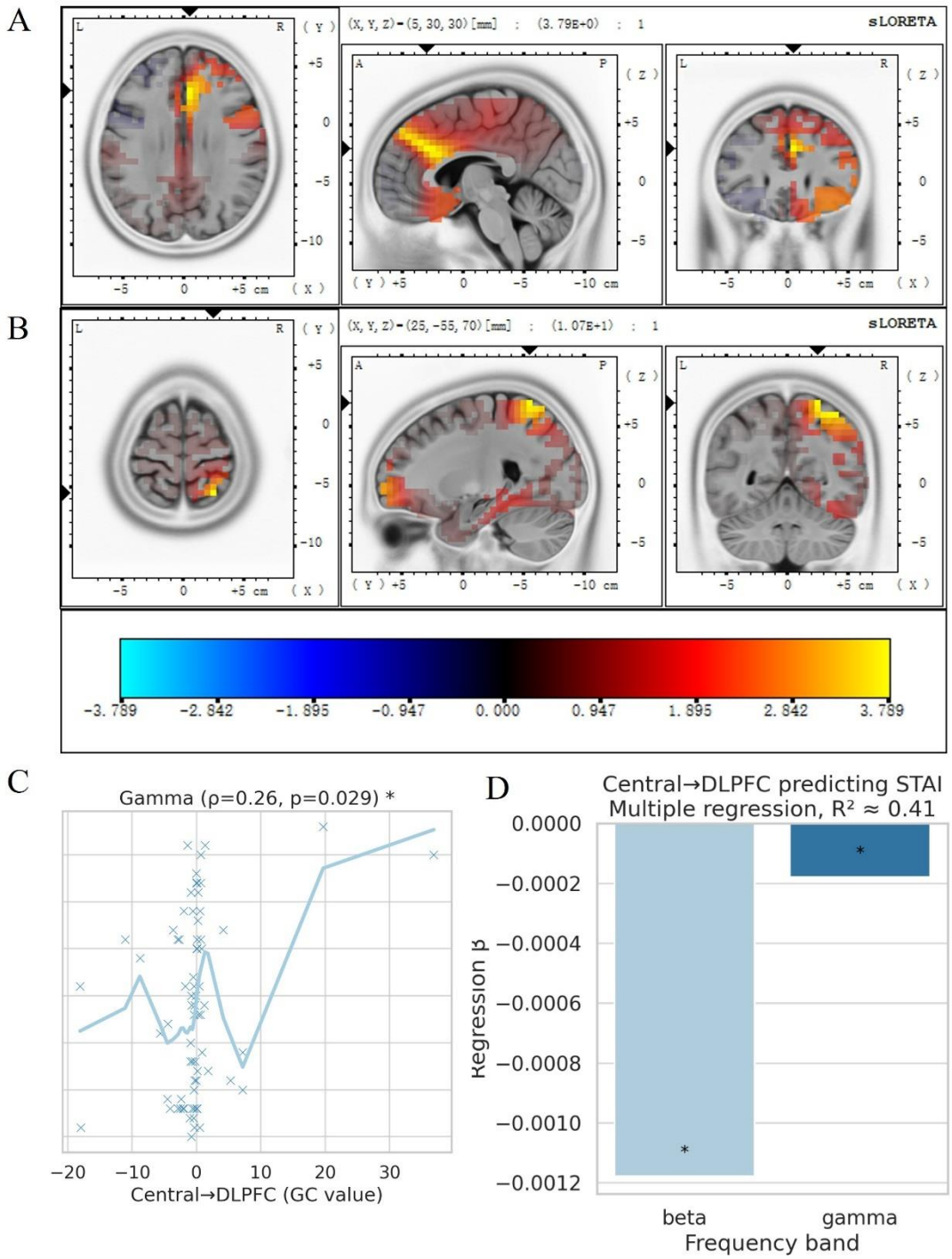

A
B
C
D
E
F
PA
PI
VA
VI
Average per 10 Hz
Amplitude (dB)
Frequency (Hz)
power density of frequency
The Roughness (a.u.)
****
piano
violin
musical element

Figure 5


A
(X, Y, Z) = (5, 30, 30) [mm] : (3.79E+0) : 1
sLORETA
B
(X, Y, Z) = (25, -55, 70) [mm] : (1.07E+1) : 1
sLORETA
-3.789 -2.842 -1.895 -0.947 0.000 0.947 1.895 2.842 3.789
C
Gamma (ρ=0.26, p=0.029) *
Central→DLPFC (GC value)
D
Central→DLPFC predicting STAI
Multiple regression, R² ≈ 0.41
Regression β
beta
gamma
Frequency band